

Week 10

Linear Discriminant Analysis

Projecting for Class Separation

Section	Question it answers
1. From Variance to Separation	Why doesn't the direction that captures the most variance always separate classes best?
2. Two Philosophies of Projection	What exactly is the difference between PCA and LDA?
3. Fisher's Criterion	How does LDA find the projection that maximises class separation?
4. Testing LDA on Our Data	How does LDA compare against the five methods from Weeks 5-9?
5. LDA as Dimensionality Reduction	What does neural population activity look like in discriminant space?
6. When Assumptions Break	What happens when healthy and impaired have different covariance?
7. From LDA to QDA	How do we fix LDA when its assumptions fail?
8. Bringing It Together	Where does LDA fit in the comparison table?

1. From Variance to Separation

Why doesn't the direction that captures the most variance always separate classes best?

Imagine a rehabilitation clinic running a brain-computer interface trial for stroke survivors. The patient has a 96-channel electrode array implanted over motor cortex. During a calibration session, they attempt reaches toward eight targets while the system records neural firing rates. The goal: build a decoder that maps neural activity to intended reach direction so the patient can control a robotic arm in real time.

The engineering team tries PCA first — the tool they have used since Week 4 of this course. They project the 96-channel recordings onto the first two principal components and colour each trial by its target direction. The result is disappointing: the eight direction clusters overlap so heavily that no classifier could separate them. The team adds more PCA components — 5, 10, 20 — but accuracy plateaus around 70%. Something is wrong.

The problem is not the data. The neurons are cosine-tuned and carry clear directional information. The problem is what PCA optimises. PCA finds the axes of maximum total variance. In this patient's recordings, the largest source of variance is not the direction signal — it is slow fluctuations in overall cortical excitability across the session. Some minutes, the entire population fires more actively; other minutes, less. These global fluctuations dominate the variance budget, so PCA devotes its first components to capturing them. The direction information — subtle differences in which subset of neurons fires for each target — is buried in later components, mixed with noise.

A colleague suggests trying LDA instead. Linear Discriminant Analysis does not care about total variance. It asks a different question: which projection makes the eight direction labels most distinguishable? LDA decomposes the total variance into two pieces — the part that differs between directions (useful signal) and the part that varies within each direction (noise, including those global excitability fluctuations). Then it finds the projection that maximises the ratio of the first to the second.

The result is dramatic. LDA's first two discriminant axes separate the eight direction clusters cleanly. Decoding accuracy jumps to 95%. The BCI works. And LDA achieved this with no hyperparameters to tune — no C, no k, no kernel, no grid search. Just a single matrix eigendecomposition, computed in milliseconds.

This scenario is not hypothetical in spirit. It mirrors a real challenge in BCI research: the mismatch between what PCA preserves (variance) and what decoders need (class separation). Kaufman et al. (2014) demonstrated in primate motor cortex that the dominant principal components capture condition-invariant activity — shared fluctuations that are the same regardless of which movement the animal is preparing — rather than condition-specific signals that distinguish reach directions. The task-relevant modulation is smaller in magnitude and buried in later components. Techniques that explicitly target the discriminative subspace — like LDA — bypass this problem entirely.

We can see this in our own course dataset. Take our 80-neuron recordings and project all 480 trials onto a single axis. If we use PCA's first principal component (PC1), the eight direction

classes overlap substantially. Now project the same 480 trials onto LDA's first discriminant axis (LD1). The eight direction clusters pull apart, because LDA explicitly maximises the separation between direction means while compressing each direction's internal spread.

Same 480 neural trials, different projections — better $\neq$ perfect

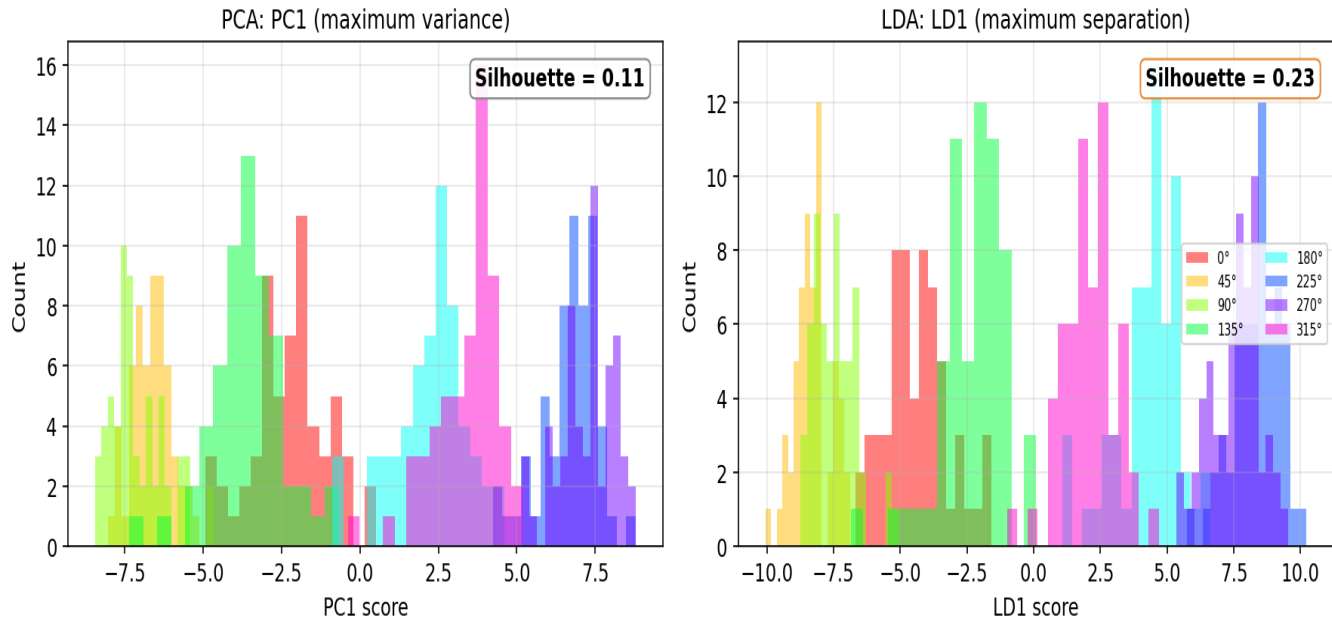


Figure 1. Same 480 neural trials projected to 1D. (A) PCA's PC1 captures maximum total variance — the eight direction distributions overlap heavily (silhouette score = 0.11). (B) LDA's LD1 maximises class separation — the distributions pull apart (silhouette = 0.23, a 2× improvement). But note the substantial residual overlap: a single discriminant axis is not enough to separate 8 directions cleanly. We will need multiple LDA components (Section 4).

The silhouette score measures, for each trial, how much closer it is to trials of its own direction compared to trials of the nearest other direction. A score of 1.0 means perfect separation; 0.0 means trials sit on the boundary between clusters. LD1 doubles the score from 0.11 to 0.23 — a real improvement — but 0.23 is still far from perfect. This makes sense: eight directions arranged around a circle cannot be collapsed onto a single line without overlap. The 0° and 180° clusters may separate along LD1, but 90° and 270° will project to similar values and need a second axis (LD2) to separate. The key point is that even one supervised axis does more useful work than one unsupervised axis. By Section 4 we will see that 2–3 LDA components recover most of the available separability.

Both projections are valid linear combinations of 80 firing rates. Both reduce the data to a single number per trial. The difference is the objective: PCA maximises total variance ($\Sigma_{\text{total}} = S_B + S_W$), while LDA maximises the ratio of between-class scatter (S_B) to within-class scatter (S_W). In the rehabilitation clinic scenario, the global excitability fluctuations inflate S_W — they are noise within each direction. PCA preserves them because they are high-variance. LDA suppresses them because they do not help distinguish directions. We will formalise this distinction in Sections 2 and 3.

2. Two Philosophies of Projection

What exactly is the difference between PCA and LDA — conceptually, mathematically, and practically?

Before we derive Fisher’s criterion, we need to build deep conceptual intuition about what PCA and LDA each optimise. Students used PCA in Week 4 as a tool for extracting muscle synergies. Now we need to understand why a supervised alternative exists and when it matters.

PCA asks: “Which direction captures the most total variance in the data?” It never looks at the labels. A PCA on our 480 reaching trials gives exactly the same result whether the trials are labelled by direction, by subject, or not labelled at all. LDA asks: “Which direction makes the class labels most distinguishable?” It decomposes total variance into useful variance (between-class scatter S_B — the part that separates directions) and nuisance variance (within-class scatter S_W — the part that is noise within each direction). Then it maximises the ratio S_B / S_W .

Why does this matter for motor control data? Our 80-neuron dataset has three major sources of variance: (1) direction-related tuning differences — the signal we care about; (2) inter-subject differences in baseline firing rates — nuisance, since different subjects have different overall activity levels for the same direction; and (3) trial-to-trial noise — nuisance, since even the same subject reaching in the same direction varies from trial to trial. PCA ranks axes by total variance, so it may put inter-subject differences first if those are the largest source of variability. LDA explicitly suppresses sources (2) and (3) because they contribute to S_W , not S_B .

Two Philosophies: PCA preserves variance, LDA preserves class separation

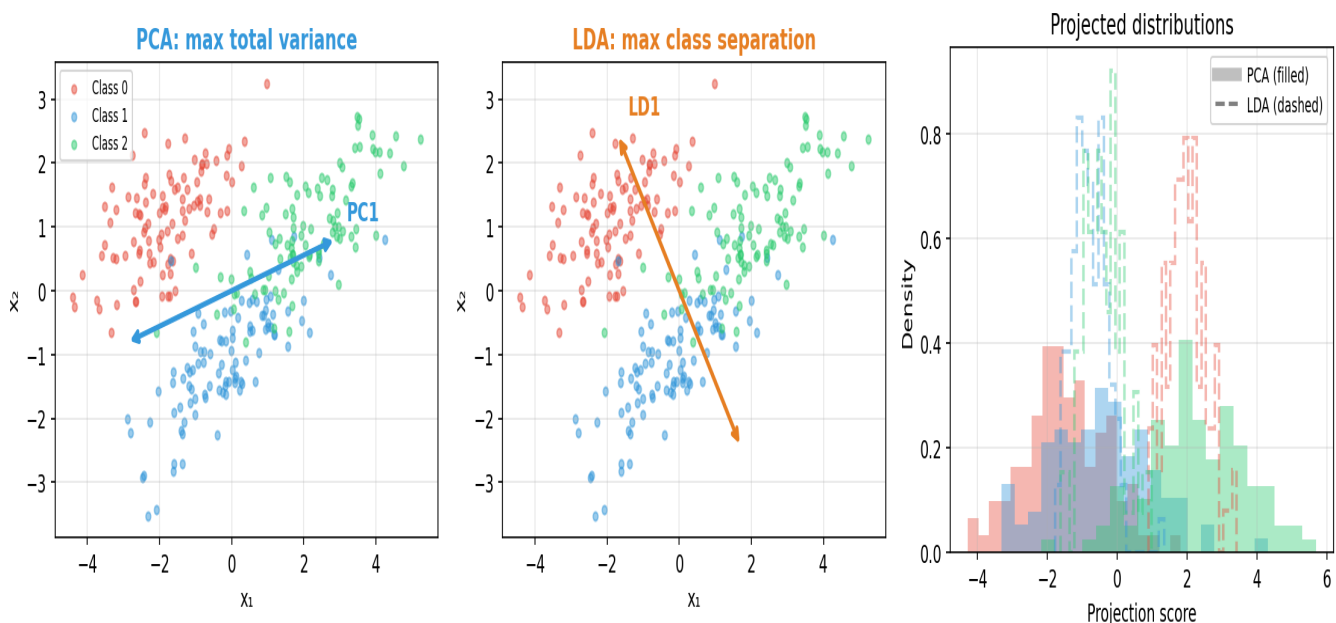


Figure 2. Two philosophies of projection on toy data with three elongated classes. Left: PCA’s PC1 aligns with the direction of maximum total variance. Centre: LDA’s LD1 aligns with the direction that best separates the three class means while compressing within-class spread. Right: the resulting 1D distributions — PCA (filled) overlaps substantially; LDA (dashed) separates the classes.

The decomposition that connects PCA and LDA

Both PCA and LDA solve eigendecomposition problems — they just decompose different matrices. To see the relationship precisely, start with the total covariance matrix that PCA decomposes. For N data points x_i with grand mean μ , the total scatter matrix is:

$$\Sigma_{\text{total}} = (1/N) \sum_i (x_i - \mu)(x_i - \mu)^T$$

This is the matrix that PCA eigendecomposes. Its eigenvectors are the principal components; its eigenvalues are the variances along each axis. PCA does not use the class labels y_i anywhere — Σ_{total} is computed from all trials pooled together.

Now suppose each trial belongs to one of K classes. We can decompose each trial's deviation from the grand mean into two parts: the deviation of the class mean from the grand mean (between-class), plus the deviation of the trial from its own class mean (within-class):

$$(x_i - \mu) = (\mu_k - \mu) + (x_i - \mu_k)$$

where μ_k is the mean of the class to which trial i belongs. Substituting this into the total scatter and expanding (the cross-term vanishes), we get:

$$\Sigma_{\text{total}} = S_B + S_W$$

where the between-class scatter captures how far apart the class centres are:

$$S_B = (1/N) \sum_k N_k (\mu_k - \mu)(\mu_k - \mu)^T$$

and the within-class scatter captures how much trials spread around their own class mean:

$$S_W = (1/N) \sum_k \sum_{i \in \text{class } k} (x_i - \mu_k)(x_i - \mu_k)^T$$

This decomposition is the key insight. PCA eigendecomposes $\Sigma_{\text{total}} = S_B + S_W$ — it finds axes that capture the most combined variance, with no way to distinguish signal from noise. LDA eigendecomposes $S_W^{-1} S_B$ — it finds axes that maximise between-class variance relative to within-class variance. If S_W is large (high noise), PCA's top axes will be dominated by noise. LDA's inverse S_W^{-1} downweights exactly those noisy directions.

For our reaching data: the direction signal lives in S_B (class means differ because neurons have different preferred directions). Inter-subject baseline differences and trial-to-trial noise live in S_W (they vary within each direction class). When we use PCA, we ask: which projection maximises $S_B + S_W$? When we use LDA, we ask: which projection maximises S_B / S_W ? These are genuinely different questions, and they give different answers whenever S_W is not proportional to S_B .

When do PCA and LDA agree?

When the class differences dominate the total variance ($S_B \gg S_W$), the first PC and the first LD point in roughly the same direction. This happens when classes are far apart relative to their internal spread and there is little nuisance variance. For our cosine-tuned neurons, the direction signal is strong, so PCA and LDA may partially agree on the top 1-2 axes. But for EMG data — where inter-subject amplitude differences and synergy correlations add large nuisance variance — LDA should produce a visibly different and better projection.

When do they disagree badly? When nuisance variance dominates ($S_W \gg S_B$). This is exactly what happens in high-dimensional neuroscience data with many recording channels and few trials per condition — the regime where LDA's supervised objective becomes essential.

Practical implications for the course so far

In Week 4, PCA on EMG found synergy axes — these captured the most variable patterns, which happened to be physiologically interpretable (extensor, flexor, elbow modules). But PCA made no promise about separating directions. In Week 5, PCA + logistic regression used PCA as preprocessing before classification. This worked because LR could learn to ignore uninformative variance dimensions. But it was inefficient: why project into a space that preserves noise, only to ask the classifier to ignore it? LDA offers a cleaner pipeline: project directly into the space that matters for the classification task.

3. Fisher's Criterion

How does LDA find the projection that maximises class separation?

We start with the two-class case: healthy versus impaired, using 6 EMG features. We want a single direction w in 6D EMG space such that projecting onto w separates the two groups as cleanly as possible. Fisher's criterion formalises "separation" as the ratio of between-class distance to within-class spread in the projected space:

$$J(w) = (\tilde{\mu}_1 - \tilde{\mu}_2)^2 / (\tilde{s}_1^2 + \tilde{s}_2^2)$$

where $\tilde{\mu}_k$ and $\tilde{s}_k^2$ are the mean and variance of class k after projection onto w . The numerator measures how far apart the class centres are; the denominator measures how spread out each class is. Maximising J means pushing the means apart while squeezing the classes tight.

This can be written in matrix form as $J(w) = w^T S_B w / w^T S_W w$, where S_B is the between-class scatter matrix (outer products of class mean differences) and S_W is the within-class scatter matrix (sum of each class's covariance). The solution is $w^* = S_W^{-1}(\mu_1 - \mu_2)$ — no iterative optimisation, no learning rate, no gradient descent. Just a matrix inverse and a subtraction.

Notice that w^* does not point from μ_1 to μ_2 . If the within-class scatter were spherical (identity covariance), it would. But when the scatter is elongated — as it is in EMG space due to muscle synergy correlations — the inverse S_W^{-1} tilts the projection direction to account for the shape of the scatter. This is exactly what Figure 3 illustrates.

Fisher's Criterion on EMG Binary Diagnosis

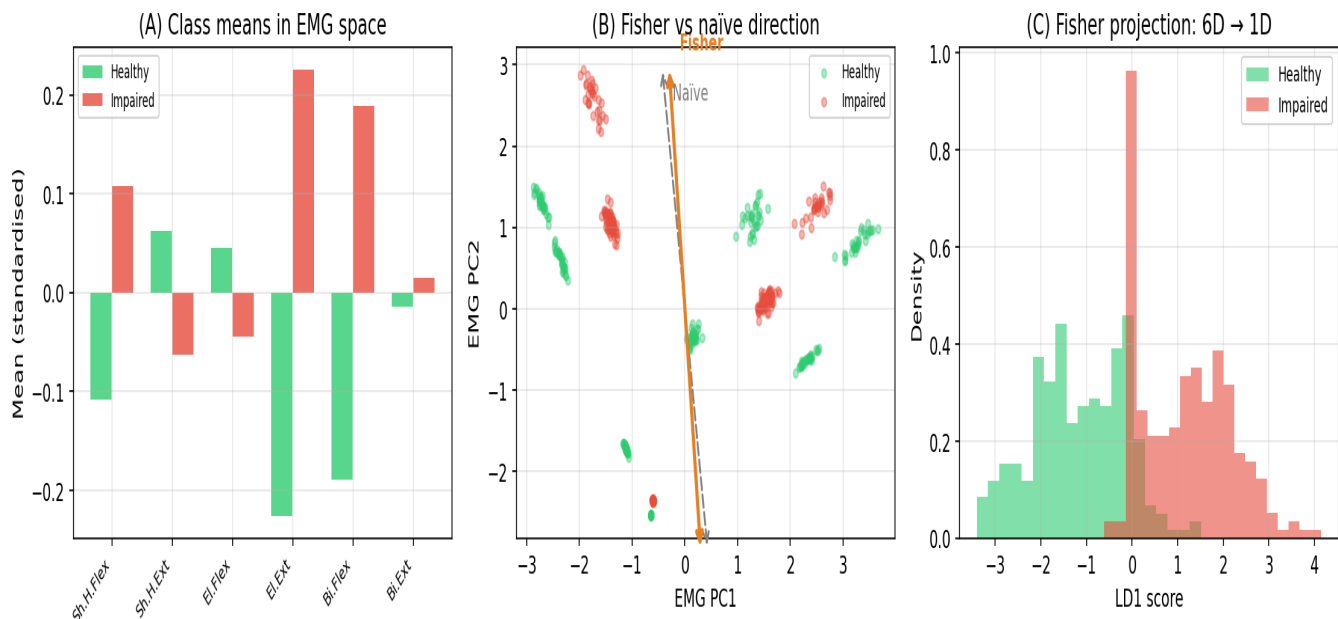


Figure 3. Fisher's criterion on EMG binary diagnosis. (A) Class means in 6D EMG space — the muscles that differ most between healthy and impaired. (B) Fisher's direction (orange) tilts away from the naive mean-connecting direction (grey) to account for the elongated within-class scatter. (C) The resulting 1D projection cleanly separates the two groups.

The multi-class generalisation

For 8 reach directions, Fisher's criterion generalises to the eigendecomposition of $S_W^{-1}S_B$. This matrix has at most $\min(K-1, D)$ nonzero eigenvalues, where K is the number of classes and D is the number of features. For our 8-direction task with 80 neurons, we get at most 7 discriminant axes — compared with 80 PCA components. LDA compresses the data more aggressively because it uses the labels to identify the relevant subspace.

Connection to Week 5 (Logistic Regression): Both LDA and LR produce linear decision boundaries. LR finds its boundary by maximising $P(y|x)$ via gradient descent; LDA finds its boundary by maximising Fisher's criterion via a closed-form eigendecomposition. When the LDA assumptions hold (Gaussian classes with equal covariance), the two methods converge to the same boundary. When the assumptions are violated, they can differ — and we will test exactly this in Sections 6 and 7.

Fisher's criterion accounts for the shape of the scatter, not just the distance between means

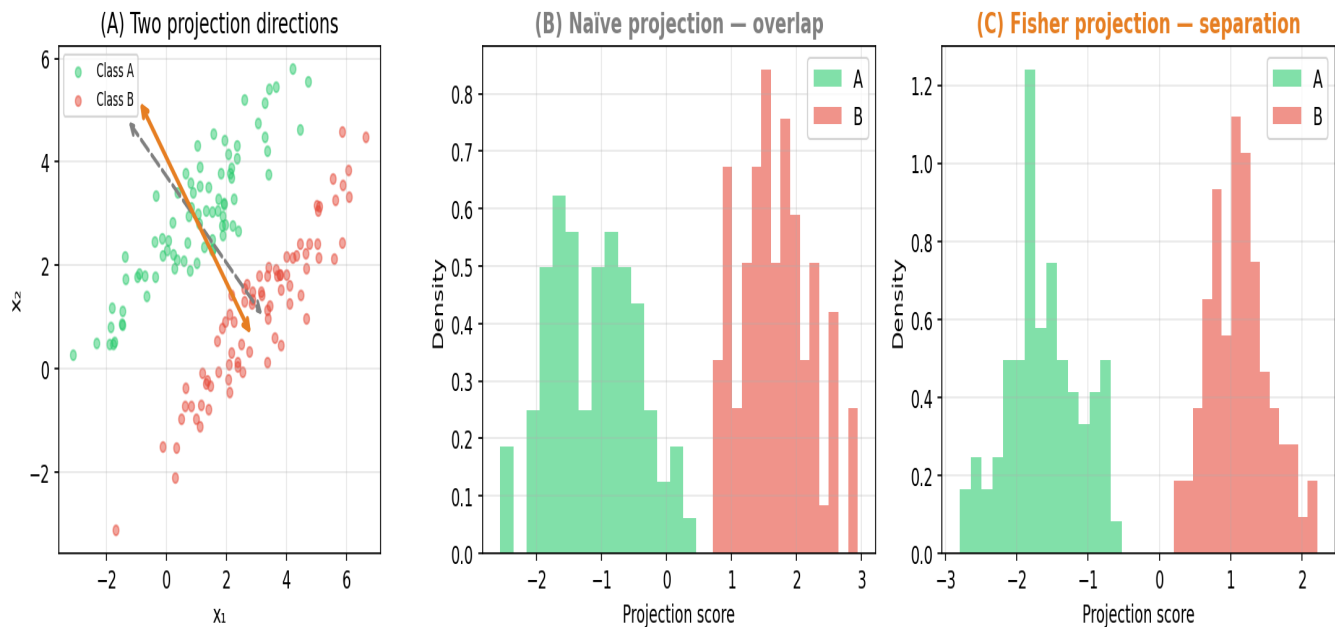


Figure 4. Geometric intuition. Two elongated classes where the naïve projection (line connecting means) gives poor separation because it projects along the long axis of each ellipse. Fisher's direction tilts perpendicular to the elongation, achieving clean separation.

4. Testing LDA on Our Data

How does LDA compare against the five methods from Weeks 5-9?

We now have six classifiers in our toolbox: Naive Bayes (Week 7), Logistic Regression (Week 5), KNN (Week 8), Linear SVM (Week 9), RBF SVM (Week 9), and LDA. We evaluate all six on the same four task-feature combinations using the same leave-one-subject-out protocol.

Six Methods Compared (LOSO) — LDA highlighted

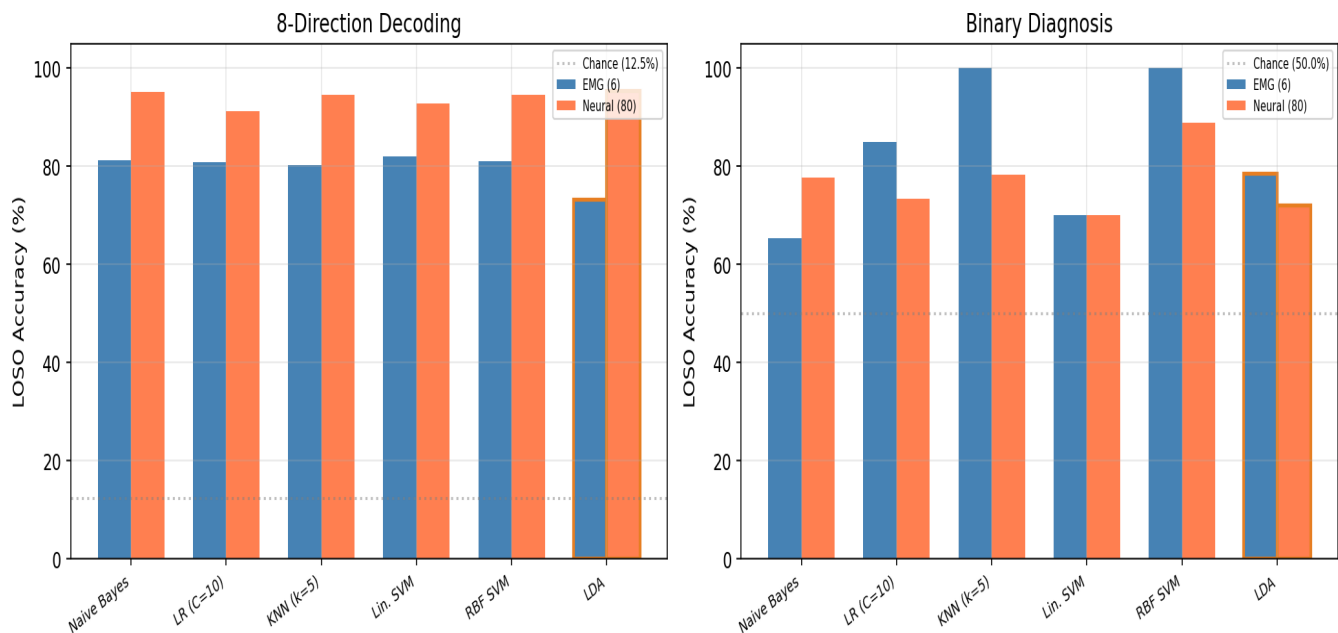


Figure 5. Six-method LOSO comparison across four tasks. LDA (highlighted with orange borders) is competitive with the best linear methods on neural direction decoding and achieves this with no hyperparameters to tune.

Several patterns emerge. On neural direction decoding, LDA achieves 95.4% — the best among all six methods, edging out Naive Bayes (95.2%) and KNN/RBF SVM (94.6%). This makes sense: cosine-tuned neurons create approximately Gaussian direction clusters with similar covariance structure, which is exactly LDA's home turf. On EMG binary diagnosis, LDA (78.5%) underperforms RBF SVM (100%) and KNN (100%). The healthy-vs-impaired distinction in EMG space involves different covariance structures — LDA's equal-covariance assumption is violated, and we will explore this in Section 6.

LDA's hyperparameter advantage

LDA has no tuning parameters. No C to sweep (SVM, LR), no k to choose (KNN), no kernel to select (SVM), no independence assumption to validate (NB). The only choice is how many discriminant components to keep — and even that has a natural maximum ($K-1 = 7$ for our 8-direction task). Compare this with the grid search burden: SVM required testing $C \times \gamma \times \text{kernel}$ in Week 9; LR required a C sweep in Week 5. LDA's closed-form solution trains in microseconds.

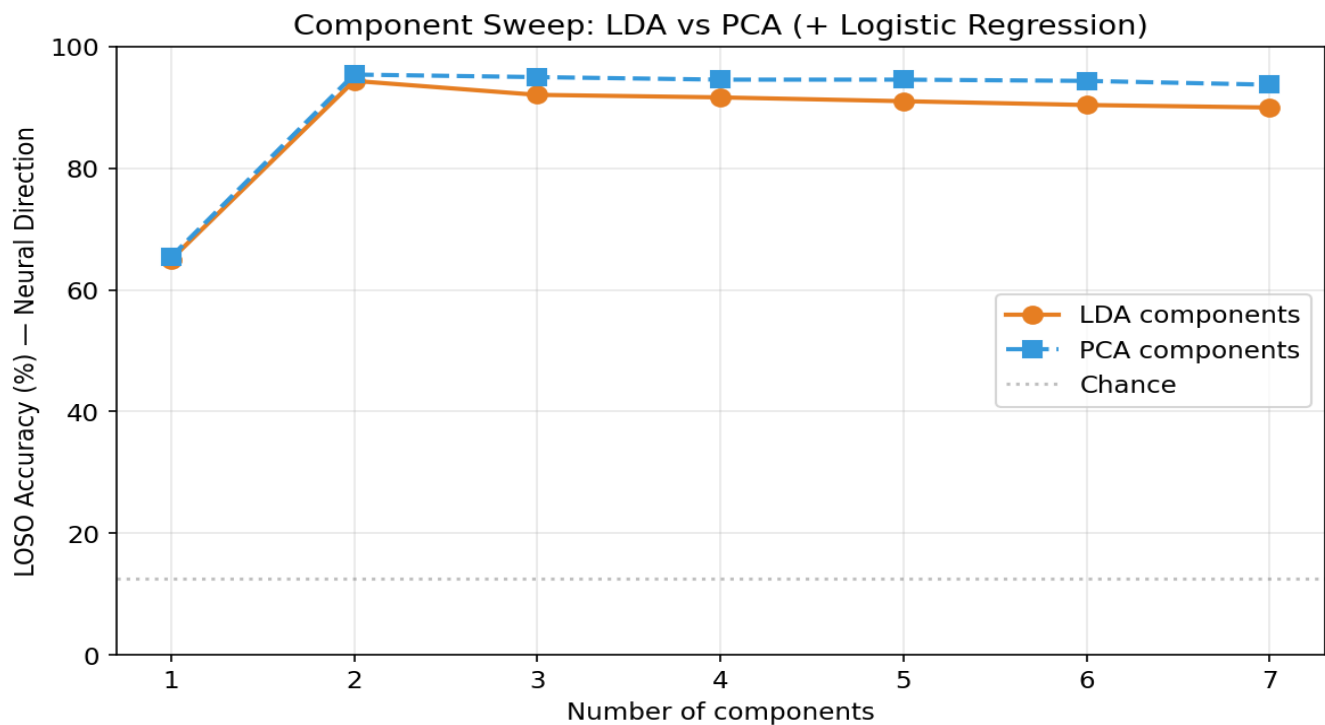


Figure 6. LO SO accuracy vs number of projection components for neural 8-direction decoding. LDA (orange) reaches near-peak accuracy with just 2–3 components out of 7 possible. PCA (blue) requires more components to reach the same level because its early components may capture nuisance variance.

5. LDA as Dimensionality Reduction

What does neural population activity look like in discriminant space?

LDA is not only a classifier — it is also a dimensionality reduction tool. For our 80-neuron dataset, LDA compresses the data from 80 features to at most 7 discriminant dimensions. This is a much more aggressive compression than PCA, which would retain as many components as there are features. The compression works because LDA targets the subspace where the class structure lives, discarding dimensions that only contribute noise.

Figure 7 compares PCA and LDA projections side by side for both feature sets. Read each row left-to-right: the left panel (PCA) and right panel (LDA) show the same data projected into 2D by different methods. The silhouette score in each panel quantifies how well the eight direction clusters separate — higher means tighter, more distinct clusters.

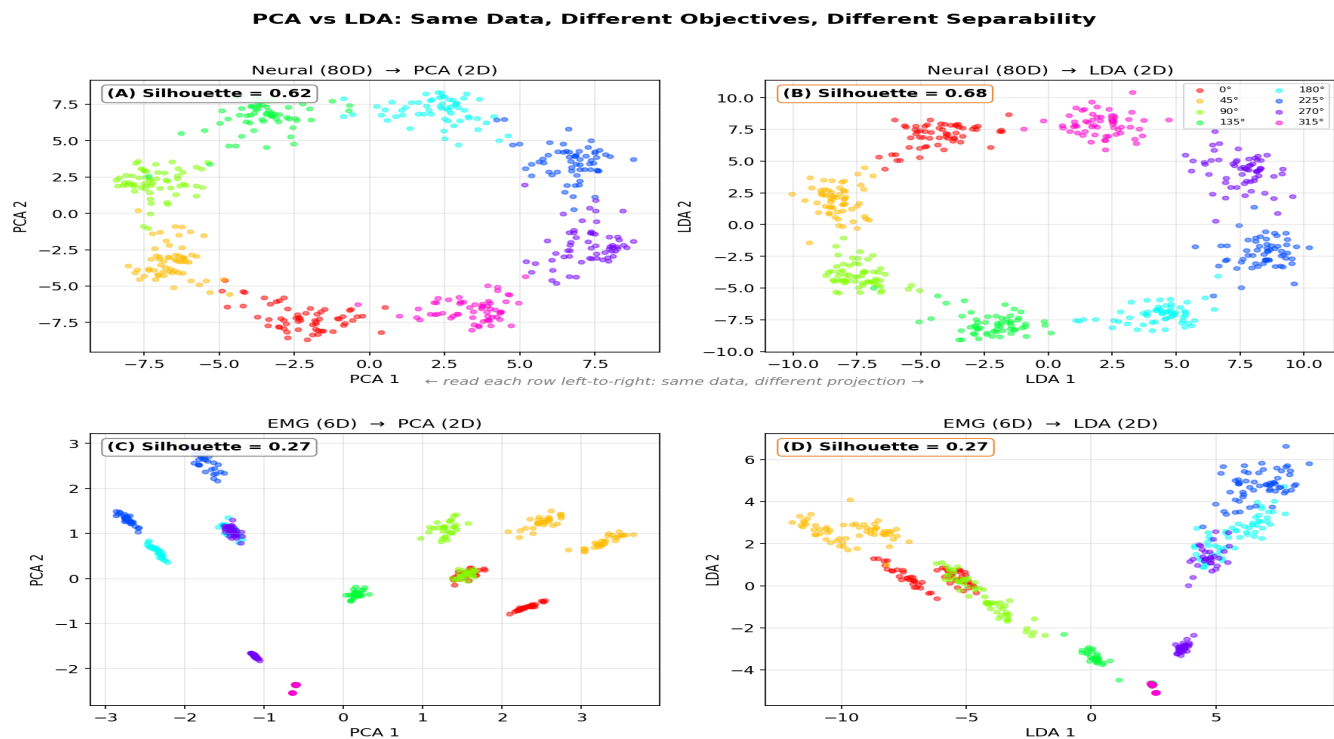


Figure 7. PCA vs LDA on both feature sets. Read each row left-to-right: same data, different projection. Silhouette scores quantify cluster separation (higher = better). (A→B) Neural data: LDA improves silhouette from 0.62 to 0.68 — clusters tighten visibly. (C→D) EMG data: silhouettes are similar (0.27 vs 0.27) — with only 6 features, PCA and LDA find comparable projections.

Top row (neural, 80D): Compare panels A and B. In PCA (A, silhouette = 0.62), the eight direction clusters are visible but overlap at their edges — inter-subject baseline differences spread each cluster along the high-variance axis. In LDA (B, silhouette = 0.68), the same eight clusters pull apart and tighten. LDA's projection explicitly suppresses the inter-subject variance that PCA preserves. The improvement is moderate (0.62 → 0.68) rather than dramatic, because cosine tuning is strong enough that the direction signal already contributes substantial variance — PCA partially captures it even without labels.

Bottom row (EMG, 6D): Compare panels C and D. With only 6 features, PCA and LDA find nearly identical projections (both silhouette $\approx$ 0.27). Why? In a 6-dimensional space, two components

already capture a large fraction of the total structure. There is less room for PCA and LDA to disagree because the space is small enough that the dominant variance directions and the dominant discrimination directions overlap. The low silhouette values (0.27) reflect the inherent difficulty of separating 8 directions from only 6 muscle recordings — the EMG feature space is simply not rich enough for clean 2D separation.

What do the discriminant axes mean?

Each discriminant axis is a weighted sum of the 80 neural firing rates. We can interpret these weights by asking: which neurons contribute most to each axis? Figure 8 plots the LD1 and LD2 loading weights across all 80 neurons, with each bar coloured by that neuron's preferred direction.

The pattern is striking: LD1 loads heavily on neurons with preferred directions near 0° and 180° (the horizontal axis), while LD2 loads on neurons near 90° and 270° (the vertical axis). LDA has rediscovered the directional structure of the motor cortex population — without being told about cosine tuning. The eigenvectors of $S_W^{-1}S_B$ naturally align with the axes of maximum directional contrast in the neural population, which are the cardinal directions of the reaching task.

LDA Rediscovered Directional Structure of the Neural Population

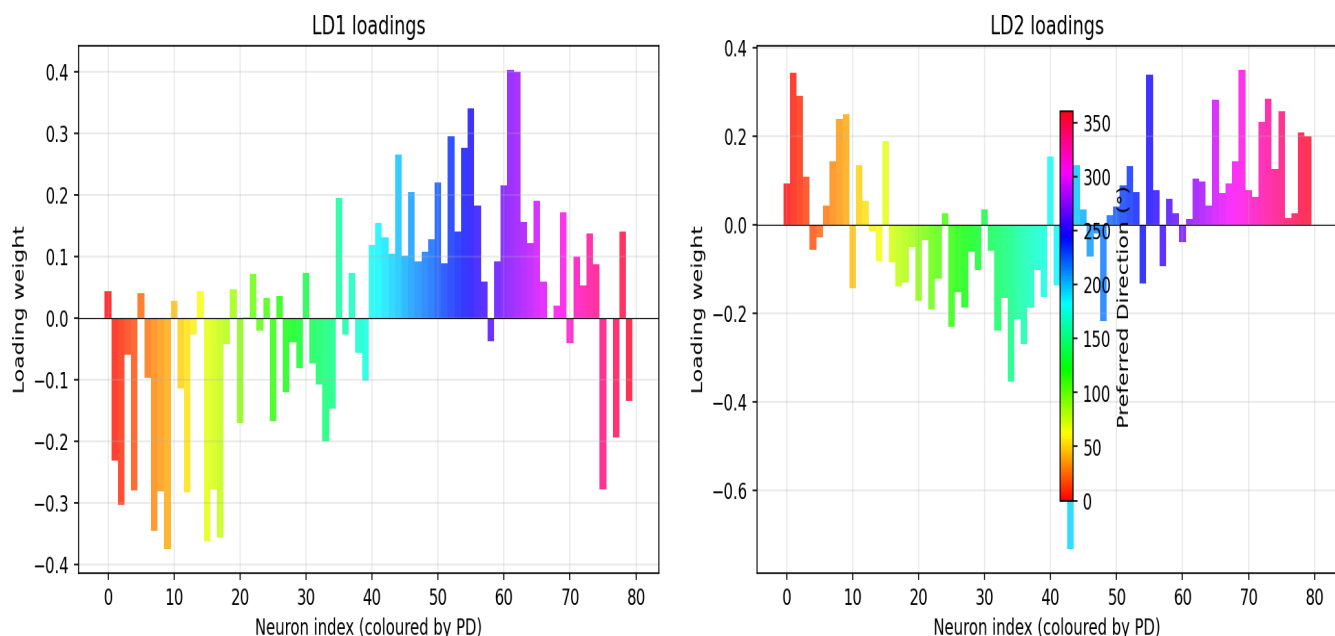


Figure 8. LDA loading vectors for the neural 8-direction task. Each bar represents one neuron's contribution to the discriminant axis, coloured by the neuron's preferred direction. LD1 weights neurons near 0°/180°; LD2 weights neurons near 90°/270°. LDA rediscovered the cosine tuning structure of Week 6 from the scatter matrices alone.

Connection to neural trajectory analysis: In motor neuroscience, researchers routinely project neural population activity into low-dimensional subspaces to visualise “neural trajectories” (Churchland et al., 2012). PCA is the default tool. LDA offers a supervised alternative when class labels are available. The choice matters: PCA trajectories show the most variable activity patterns; LDA trajectories show the patterns that best distinguish experimental conditions. For our reaching data, the LDA trajectory would trace a circle through the eight direction clusters —

the same circular structure we saw with the population vector in Week 6, but now extracted by an eigendecomposition rather than by assuming cosine tuning.

6. When Assumptions Break

What happens when healthy and impaired subjects have different covariance structures?

LDA's core assumption is that all classes share the same within-class covariance matrix S_W . This is the equal covariance (homoscedasticity) assumption. It means that regardless of whether a trial is 0° or 180° , healthy or impaired, the spread of trials around the class mean should have the same shape and size. If this holds, the optimal boundary is linear. If it fails, LDA's pooled S_W misrepresents the actual class geometry.

We can test this directly on our data. Figure 9A shows the covariance ellipses for the 8 directions in healthy subjects only. The ellipses are roughly similar in size and orientation — cosine tuning creates comparable scatter in all directions. The equal-covariance assumption is approximately satisfied. This explains LDA's strong performance on neural direction decoding (95.4%).

Figure 9B tells a different story. Healthy subjects (green ellipse) and impaired subjects (red ellipse) have visibly different covariance structures. Impaired subjects have merged muscle synergies (Week 4), which changes the correlation structure of their neural activity. The equal-covariance assumption is violated. LDA's pooled covariance averages the two different structures, misrepresenting both. This explains LDA's weaker performance on binary diagnosis (78.5% vs RBF SVM's 100%).

Testing LDA's Equal-Covariance Assumption on Our Data

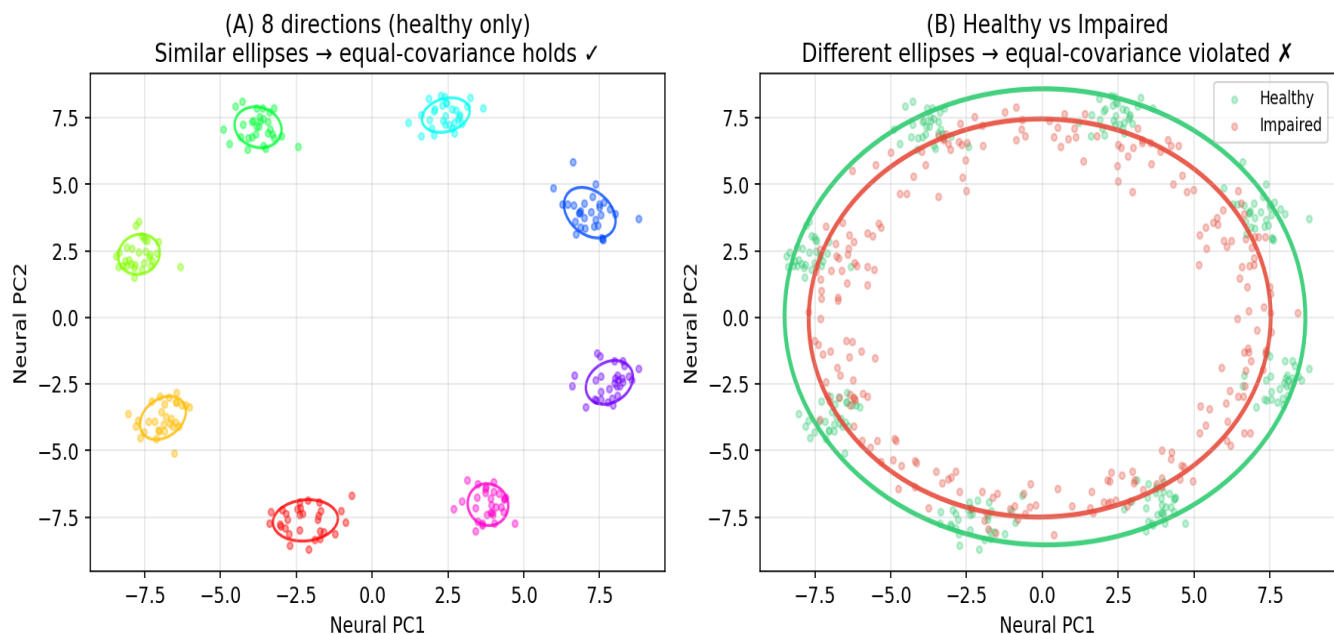


Figure 9. Testing LDA's equal-covariance assumption. (A) Eight reach directions in healthy subjects: similar ellipses — assumption holds. (B) Healthy vs impaired: different ellipses — assumption violated. Synergy merging changes the motor variability structure.

Connection to Week 7 (Naive Bayes): In Week 7, we saw that NB's independence assumption broke on EMG data because muscles are correlated through synergies. LDA and NB are both Gaussian-based classifiers with structural assumptions about the data. NB assumes diagonal

covariance (feature independence); LDA assumes equal full covariance across classes. NB's assumption breaks when features are correlated; LDA's breaks when groups have different covariance. In Week 7, PCA rescued NB by decorrelating the features. What rescues LDA? We address this next.

7. From LDA to QDA and Regularised Variants

How do we fix LDA when its assumptions fail?

The natural fix is to allow each class to have its own covariance matrix Σ_k . This is Quadratic Discriminant Analysis (QDA). The decision boundary becomes quadratic (curved) instead of linear, which can better capture the geometry of classes with different scatter.

The cost is data hunger. LDA estimates one pooled covariance matrix; QDA estimates K separate matrices. For our EMG binary task (6 features, 240 trials per class), this is manageable: $6 \times 6 = 36$ parameters per covariance matrix, with plenty of data. For our neural 8-direction task (80 features, 60 trials per direction), it is dangerous: $80 \times 80 = 6,400$ parameters per covariance matrix, estimated from only 60 trials. The covariance estimates become unreliable, and QDA overfits.

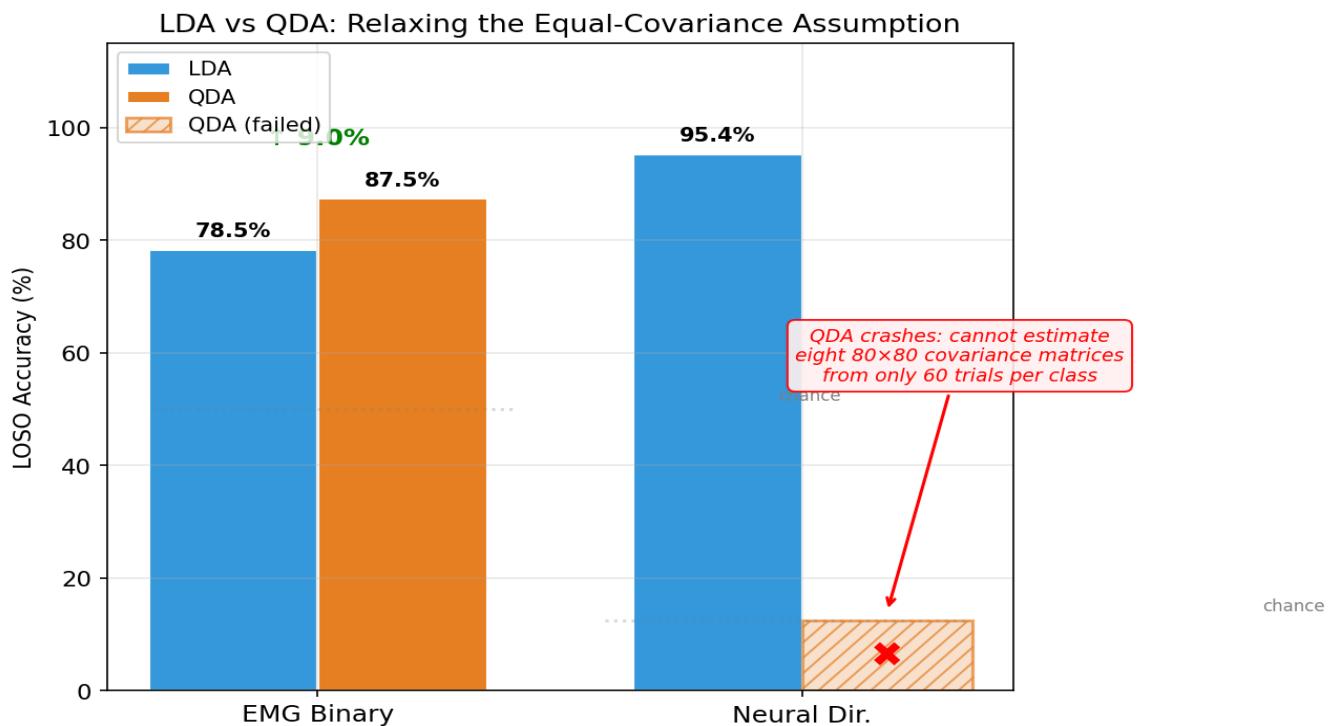


Figure 10. LDA vs QDA. On EMG binary diagnosis (left), QDA improves over LDA because it models the different covariance structures of healthy and impaired subjects. On neural 8-direction decoding (right), QDA cannot even produce a result — the hatched bar marks its failure.

The left pair tells the success story. EMG binary diagnosis has 2 classes, 6 features, and 240 trials per class. QDA needs to estimate two 6×6 covariance matrices — that is 21 unique parameters each (a symmetric 6×6 matrix has $6 \times 7 / 2 = 21$ free entries), comfortably estimated from 240 trials. QDA's accuracy jumps from 78.5% to 87.5% because it correctly models the different scatter of healthy and impaired subjects that we saw in Figure 9B.

The right pair tells the failure story. Neural 8-direction decoding has 8 classes, 80 features, and only 60 trials per class. QDA needs to estimate eight separate 80×80 covariance matrices — 3,240 unique parameters each, from just 60 trials. This is impossible: you cannot reliably estimate 3,240 numbers from 60 data points. The resulting covariance matrices are singular (not invertible), and QDA crashes entirely. Even with regularisation (`reg_param = 0.01`), the estimates

are so noisy that classification is worse than chance. LDA's pooled covariance, estimated from all 480 trials, remains stable at 95.4%.

This is a concrete instance of the bias-variance tradeoff at the model level. LDA is more biased (it assumes all classes share the same covariance) but has lower variance (it pools all the data into one estimate). QDA is less biased (it allows each class its own covariance) but has much higher variance (each estimate uses only 1/8 of the data). When the data is plentiful relative to the model complexity (EMG: 240 trials, 21 parameters), QDA's flexibility wins. When data is scarce relative to complexity (neural: 60 trials, 3,240 parameters), LDA's constraint wins.

Regularised LDA: the middle ground

Between LDA (one shared covariance) and QDA (K separate covariances) lies regularised LDA. The Ledoit-Wolf shrinkage estimator blends the sample covariance toward a structured target (diagonal or identity matrix), controlling how much the estimated covariance can deviate from a simple structure. In scikit-learn, this is `LinearDiscriminantAnalysis(solver='lsqr', shrinkage='auto')`.

This connects to a regularisation theme running through the entire course. Week 5 introduced L1/L2 regularisation for logistic regression (the C parameter). Week 9 introduced the soft margin for SVM (also controlled by C). Now Week 10 introduces covariance shrinkage for LDA. The underlying principle is the same: trade a little bias for a lot of variance reduction. In high-dimensional settings where D is close to N (more features than trials per class), some form of regularisation is essential.

8. Bringing It Together

Where does LDA fit in the comparison table?

Let us compile the full picture: six classifiers evaluated on the same four tasks, the same LOSO protocol, and now the same timing and robustness benchmarks.

Computational cost

LDA's closed-form solution (eigendecomposition of $S_W^{-1}S_B$) makes it one of the fastest methods to train. There is no iterative optimisation, no convergence criterion, no learning rate. On our 480×80 neural dataset, LDA trains in under a millisecond. Prediction is also fast: project the new trial onto the discriminant axes and assign the nearest class.

Computational Cost: LDA's Closed-Form Advantage

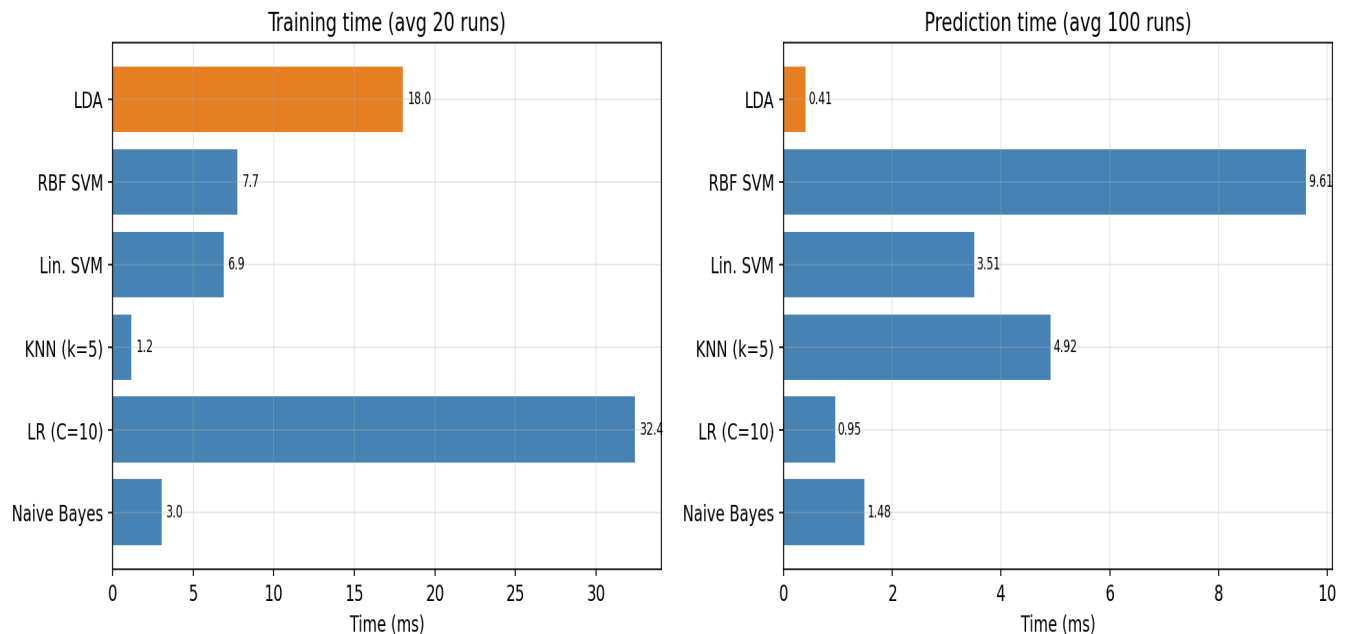


Figure 11. Training and prediction time for all six methods on the neural dataset. LDA (orange) is among the fastest, with closed-form training and simple projection for prediction.

Robustness to electrode drift

We repeat the electrode drift simulation from Week 9: add Gaussian noise of increasing standard deviation to test features without retraining the model. LDA's linear boundary responds to drift similarly to other linear methods (LR, linear SVM). Its robustness depends on whether the drift changes only the mean activity (location shift, which LDA handles well) or also the covariance structure (which invalidates the pooled S_W).

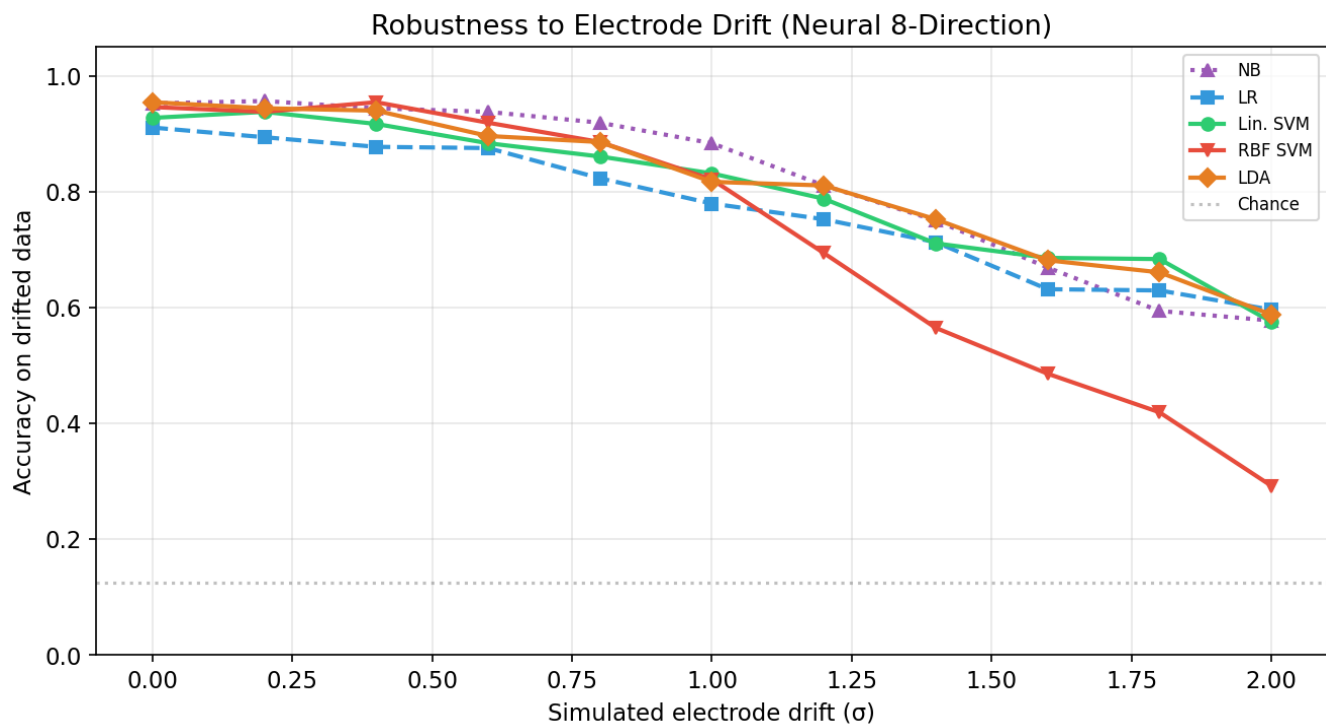


Figure 12. Accuracy under simulated electrode drift (neural 8-direction). LDA degrades at a rate comparable to other linear methods. The RBF SVM remains the most sensitive to drift due to its curved boundaries.

The updated comparison

Summary Comparison — All Six Classifiers (LOSO, Neural + EMG)

Method	EMG Dir.	Neur. Dir.	EMG Bin.	Neur. Bin.	Train (ms)	Predict (ms)	Hyperparams
Naive Bayes	81.2%	95.2%	65.4%	77.7%	3.0	1.48	None
LR (C=10)	80.8%	91.2%	85.0%	73.3%	32.4	0.95	C
KNN (k=5)	80.2%	94.6%	100.0%	78.3%	1.2	4.92	k, metric
Lin. SVM	82.1%	92.7%	70.0%	70.0%	6.9	3.51	C
RBF SVM	81.0%	94.6%	100.0%	89.0%	7.7	9.61	C, γ , kernel
LDA	73.1%	95.4%	78.5%	72.1%	18.0	0.41	None

Figure 13. Summary comparison table for all six classifiers through Week 10.

When to use LDA

LDA excels when: (1) classes are approximately Gaussian with similar covariance — which holds for cosine-tuned neural populations decoding reach direction; (2) you have no hyperparameters to tune and need a fast, reliable baseline; (3) you want to simultaneously classify and visualise data in a low-dimensional discriminant space. LDA is weakest when classes have different covariance structures (use QDA or RBF SVM) or when the decision boundary is fundamentally nonlinear (use RBF SVM or KNN).

What we learned this week

- Fisher's criterion maximises the ratio of between-class scatter to within-class scatter — it finds the projection where class means are far apart and class spreads are small.
- LDA reduces 80 neural features to at most 7 discriminant dimensions ($K-1$ for K classes), and 2–3 components capture most of the separability.
- PCA and LDA can give very different projections of the same data: PCA preserves variance (including nuisance variance), LDA preserves class separation.
- LDA's equal-covariance assumption holds for direction decoding (cosine-tuned neurons scatter similarly in all directions) but fails for healthy-vs-impaired diagnosis.
- QDA relaxes the equal-covariance assumption but requires more data — with 80 neurons and 60 trials per class, QDA overfits while LDA stays stable.
- Covariance shrinkage (Ledoit-Wolf) regularises LDA the same way C regularises LR and SVM — trading a little bias for variance reduction.

Looking ahead: Next week — Week 11: Random Forests & Ensemble Methods. Every method so far has been a single model making one prediction. What if we combined many weak learners into something stronger than any individual classifier?

References

Kaufman MT, Churchland MM, Ryu SI, Shenoy KV. Cortical activity in the null space: permitting preparation without movement. *Nature Neuroscience*, 17(3), 440–448, 2014.

Churchland MM, Cunningham JP, Kaufman MT, Foster JD, Nuyujukian P, Ryu SI, Shenoy KV. Neural population dynamics during reaching. *Nature*, 487(7405), 51–56, 2012.

Gallego JA, Perich MG, Miller LE, Solla SA. Neural manifolds for the control of movement. *Neuron*, 94(5), 978–984, 2017.

Georgopoulos AP, Kalaska JF, Caminiti R, Massey JT. On the relations between the direction of two-dimensional arm movements and cell discharge in primate motor cortex. *Journal of Neuroscience*, 2(11), 1527–1537, 1982.

Tresch MC, Saltiel P, Bizzi E. The construction of movement by the spinal cord. *Nature Neuroscience*, 2(2), 162–167, 1999.

Clark DJ, Ting LH, Zajac FE, Neptune RR, Kautz SA. Merging of healthy motor modules predicts reduced locomotor performance and muscle coordination complexity post-stroke. *Journal of Neurophysiology*, 103(2), 844–857, 2010.